



MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

COURSE PLAN

Department	:	MATHEMATICS
Course Name & code	:	Engineering Mathematics - III & MAT2153.
Semester & branch	:	III Sem B.Tech & Common to CHEMICAL and BioTech
Name of the faculty	:	Dr. Smitha Ganesh Bhat
No of contact hours/week:		3 0 0 3

ASSESSMENT PLAN

Course Outcomes (COs)

<i>At the end of this course, the student should be able to:</i>		No. of Contact Hours	Marks
CO1:	Understand and apply the concepts of Fourier Series, Fourier transforms & their properties	10	14
CO2:	Understand and apply analyticity of a complex functions and its properties.	5	6
CO3:	Understand and apply the concepts of contour integration	7	7
CO4:	Understand the concept of vector differential and integral calculus & their properties	12	17
CO5:	Apply the concepts of linear PDEs, to solve One dimension Heat and Wave equation by different methods.	6	6
Total		40	50

Components	Surprise Quizzes	Sessional Tests	End Semester/ Make-up Examination
Duration	20 to 30 minutes	60 minutes	180 minutes
Weightage	20 % (4 X 5 marks)	30 % (2 X 15 Marks)	50 % (1 X 50 Marks)
Typology of Questions	Understanding/ Comprehension; Application; Analysis; Synthesis; Evaluation	Knowledge/ Recall; Understanding/ Comprehension; Application	Understanding/ Comprehension; Application; Analysis; Synthesis; Evaluation
Pattern	Answer one randomly selected question from the problem sheet (Students can refer their class notes)	MCQ: 10 questions (0.5 marks) Short Answers: 5 questions (2 marks)	Answer all 5 full questions of 10 marks each. Each question may have 2 to 3 parts of 3/4/5/6/7 marks
Schedule	4, 7, 10, and 13 th week of academic calendar	Calendared activity	Calendared activity
Topics Covered	Quiz 1 (L 1-10 & T _{y1-y2}) (CO 1)	Test 1 (L 1-16 & T _{b1-b2}) (CO 1&2)	Comprehensive examination covering full syllabus. Students are expected to answer all questions (CO 1-5)
	Quiz 2 (L 11-21 & T _{y3-y4}) (CO 2 &3)		
	Quiz 3 (L 22-30 & T _{y5-y6}) (CO 3 &4)	Test 2 (L 17-25 & T _{b3-b4}) (CO 3&4)	
	Quiz 4 (L 31-36 & T _{y7-y8}) (CO 5)		

Course Plan

L. No./ T. No.	Topics	Course Outcome Addressed
L0	Introduction to the Course	CO
L1	Periodic functions, Even & odd functions and their properties. List of some standard integrals (definite integrals)	CO1
L2	Trigonometric or Fourier Series. Dirichlet's conditions. Euler's formulae for Fourier coefficients	CO1
L3	Problems on Fourier series of functions with period π	CO1
L4	Tutorial	CO1
L5	Fourier series expansion of functions with arbitrary period and related problems.	CO1
L6	Fourier series expansion of odd/even functions, Half range expansions	CO1
L7	Harmonic analysis, Introduction to Fourier transform	CO1
L8	Tutorial	CO1
L9	Parseval's identity and applications	CO1
L10	Fourier sine and cosine transforms, basic properties and Self reciprocal functions	CO1

L11	Tutorial	CO2
L12	Functions of complex variable. Basic concepts on limit, continuity, differentiability and analytic functions	CO2
L13	C-R equations in Cartesian and Polar co-ordinates and related problems	CO2
L14	Harmonic functions, properties of analytic functions and related problem	CO2
L15	Finding analytic function whose real or imaginary part is given	CO2
L16	Tutorial	CO3
L17	Line integrals of complex functions and related problems. Cauchy's integral theorem, Cauchy's integral formula and applications, related problems	CO3
L18	Basic concepts of power series, radius/ region of convergence. Taylor's theorem	CO3
L19	Taylor's series expansions of analytic	CO3
L20	Tutorials	CO3
L21	Laurent series expansions and related problems.	CO3
L22	Residue, Cauchy's residue theorem and related problems	CO3
L23	Tutorials	CO4
L24	Basic concepts on vectors & their products. Ordinary / partial derivatives of a vector function and related formulas.	CO4
L25	Vector differential operator, gradient, divergence and curl, Directional derivatives and their geometrical interpolation and related problems	CO4
L26	Identities involving the operator, Solenoidal / Irrotational fields and related problems	CO4
L27	Tutorials	CO4
L28	Line integrals, conservative vector field, independent of paths and related problems.	CO4
L29	Green's theorem in the plane	CO4
L30	Applications of Green's theorem and related problems	CO4
L31	Tutorial	CO4
L32	Surface / Volume integrals, related problems	CO4
L33	Stoke's theorem and Divergence theorem	CO4
L34	Tutorials	CO4
L35	Basic concepts of PDEs / Solutions. Solutions of PDEs by direct integration and by separation of variables.	CO5
L36	Solution of PDE by indicated transformations.	CO5
L37	Derivation of one dimensional wave equation. D'Alembert's solution	CO5
L38	Solution of one dimensional wave equation by separation of variables.	CO5

L39	Derivation of one dimensional heat equation and its solution by separation of variable method	CO5
L40	Tutorials	CO5

References:

1. Erwin Kreyszig: Advanced Engineering Mathematics-, Wiley Eastern
2. Grewal B.S. - Higher Engineering Mathematics, Khanna Publishers
3. Murray R. Spiegel: Vector Analysis, Schaum Publishing Co.
4. Advanced Engineering Mathematics, Vol 3, by Narayanan, Ramaniah and Manicavachagam Pillay
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Submitted by: Dr. Smitha Ganesh Bhat

(Signature of the faculty)

Date: 27-08-2022

Approved by: Dr. Sudhakara G

(Signature of HOD)

Date: Click or tap to enter a date.

FACULTY MEMBERS TEACHING THE COURSE (IF MULTIPLE SECTIONS EXIST):

FACULTY	SECTION	FACULTY	SECTION
Dr. Smitha G Bhat	Chem A		
Dr. Vidya H C	Chem B		
Dr. Swati S Nayak	BioTech		
